

1.

One of my roommates, Jonathan, loves glizzies. Unfortunately, far too often, he misplaces his glizzies somewhere in the apartment. Now, the glizzy gobbler managed to lose his glizzies again, and he needs to find them. Now, he only needs one pack of glizzies, but somehow he's managed to lose his entire supply of three glizzies! Further, he seems to have lost his tongs! He needs to find both his tongs and at least one glizzy packet, and he won't be able to pick up the glizzies until he's found the tongs because he doesn't want to pick up the raw glizzies with his bare hands! They seem to have been lost in some especially dark corner of the apartment this time. Jonathan cannot find them on his own using his antediluvian searching techniques, so we must help Jonathan find them by communicating a traditional searching algorithm to him by yelling at him from across the apartment. Hopefully, Jonny will be able to prepare his glizzies with our help.

2.

- a. States are defined as various locations Jonathan can be in within the apartment, illustrated as a square grid composed of locations where he can stand and edges that he cannot pass through.
- b. The initial state is where Jonny starts in the search algorithm—in his bedroom. At this time, he will have no glizzies and no tongs in his possession.
- c. The only actions Jonny can make are moving into an adjacent square not blocked off by an edge. The successor function will change the state into one where Jonny has moved to the chosen valid location.
- d. Does Jonathan occupy the same grid tile as one of the glizzies? Has he found the tongs yet? Both need to be true concurrently.
- e. The cost function will be how many steps it takes for him to optimally find the glizzies and tongs (so how many tiles are traversed in the search functions found path)
- f. Grid restrictions: he has to follow the layout of the apartment (cannot walk through walls).

Both objectives are combined into the cost function. Effectively, we are not including the traversal to old tiles in the illustration or step count, so if he's already found glizzies and locates the tongs after it, it will functionally win, but within the program, we need to make him remember the specific tile he found the glizzies at, so he can find them again and not forget about them.